

EDA on Diwali Dataset

```
In [1]: print("EDA Using Python Library")
```

EDA Using Python Library

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv("Diwali Sales Data.csv", encoding="latin1")
```

```
In [4]: df.shape
```

```
Out[4]: (11251, 15)
```

```
In [5]: df.columns
```

```
Out[5]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
              'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
              'Orders', 'Amount', 'Status', 'unnamed1'],
              dtype='str')
```

```
In [6]: df.head()
```

```
Out[6]:
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Categ
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	A
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	A
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	A
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	A
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	A

```
In [7]: df.tail()
```

```
Out[7]:
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Cat
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	Vete
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	

```
In [8]: df.sample(5)
```

Out[8]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status		State	Zone	Occupation	Product_Catego
4619	1004656	Häberlin	P00001142	F	26-35	34	0		Uttar Pradesh	Central	Banking	Dec
558	1003626	Kartik	P00120442	F	26-35	27	1		Uttar Pradesh	Central	Govt	Fo
9764	1003920	Sharda	P00237642	F	36-45	44	0		Delhi	Central	IT Sector	Household itei
9186	1005787	Aditya	P00120642	F	26-35	31	0		Delhi	Central	IT Sector	Footwear & Shc
3151	1001397	Ferguson	P00260042	F	36-45	37	1		Maharashtra	Western	Hospitality	Furnitu

In [9]: df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   User_ID                11251 non-null  int64  
1   Cust_name              11251 non-null  str     
2   Product_ID             11251 non-null  str     
3   Gender                 11251 non-null  str     
4   Age Group              11251 non-null  str     
5   Age                    11251 non-null  int64  
6   Marital_Status         11251 non-null  int64  
7   State                  11251 non-null  str     
8   Zone                   11251 non-null  str     
9   Occupation              11251 non-null  str     
10  Product_Category       11251 non-null  str     
11  Orders                  11251 non-null  int64  
12  Amount                  11239 non-null  float64 
13  Status                  0 non-null      float64 
14  unnamed1                0 non-null      float64 
dtypes: float64(3), int64(4), str(8)
memory usage: 1.3 MB
```

Dropping unnecessary Columns

```
In [10]: df.drop(columns=['Status', 'unnamed1'], inplace=True)
```

```
In [11]: df
```

```
Out[11]:
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	
...	...	...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	

11251 rows × 13 columns

Null values checking

```
In [12]: df.isnull().sum()
```

```
Out[12]: User_ID      0
         Cust_name    0
         Product_ID   0
         Gender        0
         Age Group     0
         Age           0
         Marital_Status 0
         State         0
         Zone          0
         Occupation    0
         Product_Category 0
         Orders        0
         Amount        12
         dtype: int64
```

```
In [13]: df[df['Amount'].isnull()]
```

Out[13]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Cate
7	1002092	Shivangi	P00273442	F	55+	61		0	Maharashtra	Western	IT Sector
14	1003858	Cano	P00293742	M	46-50	46		1	Madhya Pradesh	Central	Hospitality
16	1005447	Amy	P00275642	F	46-50	48		1	Andhra Pradesh	Southern	IT Sector
109	1005265	Sakshi	P00296242	F	46-50	48		1	Delhi	Central	Banking
111	1005261	Apoorva	P00057942	F	36-45	41		1	Delhi	Central	IT Sector
184	1005538	Kartik	P00269542	F	46-50	49		1	Karnataka	Southern	Banking
293	1000326	Jonathan	P00120542	M	51-55	53		0	Gujarat	Western	IT Sector
344	1002507	Lakshmi	P00045842	F	26-35	35		1	Gujarat	Western	Chemical
345	1004498	Srishti	P00030842	F	51-55	55		0	Delhi	Central	Textile
452	1004601	Gaurav	P00014442	F	36-45	40		1	Madhya Pradesh	Central	Hospitality
464	1004528	Anurag	P00338442	F	26-35	33		1	Uttar Pradesh	Central	Automobile
493	1002994	Hemant	P0009942	F	36-45	38		0	Uttar Pradesh	Central	IT Sector

We drop rows which have null amount

In [14]: `df.dropna(inplace=True)`

Changing Amount data type from float to int.

In [15]: `df['Amount'] = df['Amount'].astype('int')`In [16]: `df['Amount'].dtype`

```
Out[16]: dtype('int64')
```

checking Distribution

```
In [17]: df.describe().T
```

```
Out[17]:
```

	count	mean	std	min	25%	50%	75%	max
User_ID	11239.0	1.003004e+06	1716.038826	1000001.0	1001491.5	1003064.0	1004426.5	1006040.0
Age	11239.0	3.541036e+01	12.753866	12.0	27.0	33.0	43.0	92.0
Marital_Status	11239.0	4.200552e-01	0.493589	0.0	0.0	0.0	1.0	1.0
Orders	11239.0	2.489634e+00	1.114967	1.0	2.0	2.0	3.0	4.0
Amount	11239.0	9.453611e+03	5222.355168	188.0	5443.0	8109.0	12675.0	23952.0

```
In [18]: df[['Age', 'Orders', 'Amount']].describe().T
```

```
Out[18]:
```

	count	mean	std	min	25%	50%	75%	max
Age	11239.0	35.410357	12.753866	12.0	27.0	33.0	43.0	92.0
Orders	11239.0	2.489634	1.114967	1.0	2.0	2.0	3.0	4.0
Amount	11239.0	9453.610553	5222.355168	188.0	5443.0	8109.0	12675.0	23952.0

EDA

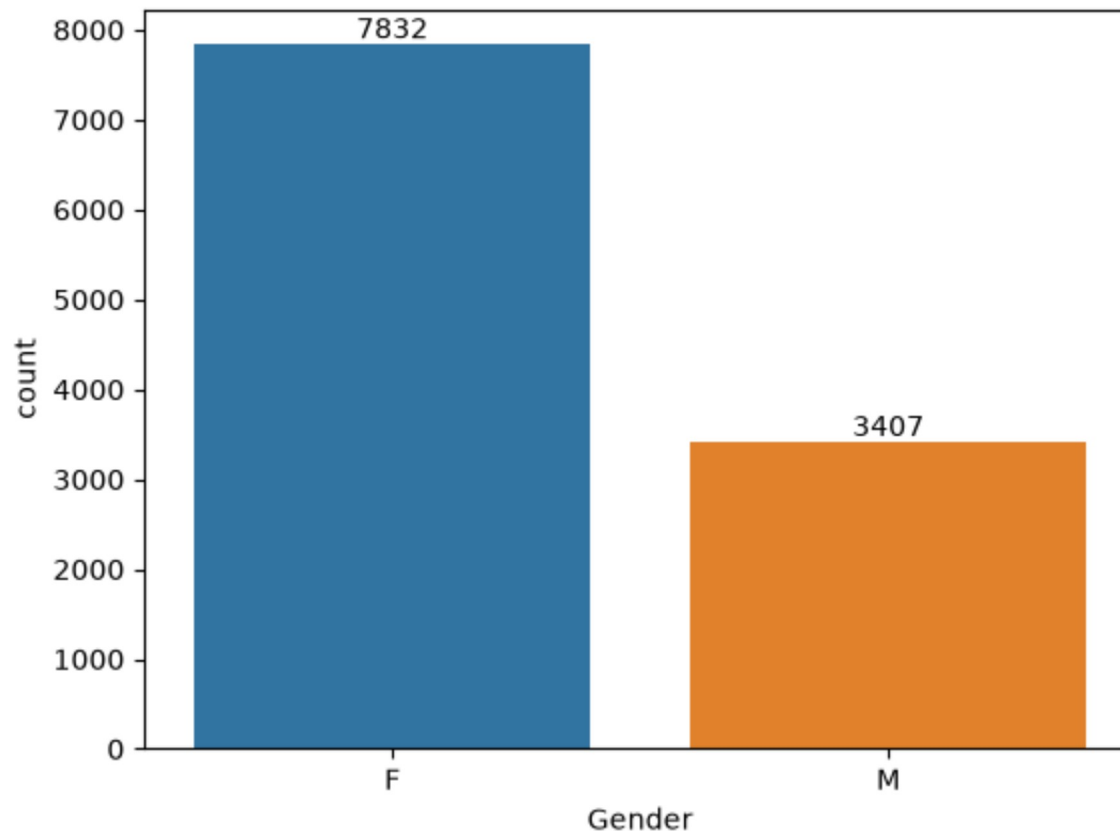
```
In [19]: df.columns
```

```
Out[19]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
               'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
               'Orders', 'Amount'],  
              dtype='str')
```

Gender wise distribution

- Insights :- Most of the buyers are females and female purchasing power is greater than males

```
In [20]: ax = sns.countplot(data=df, x='Gender', hue='Gender')
for bars in ax.containers:
    ax.bar_label(bars)
```



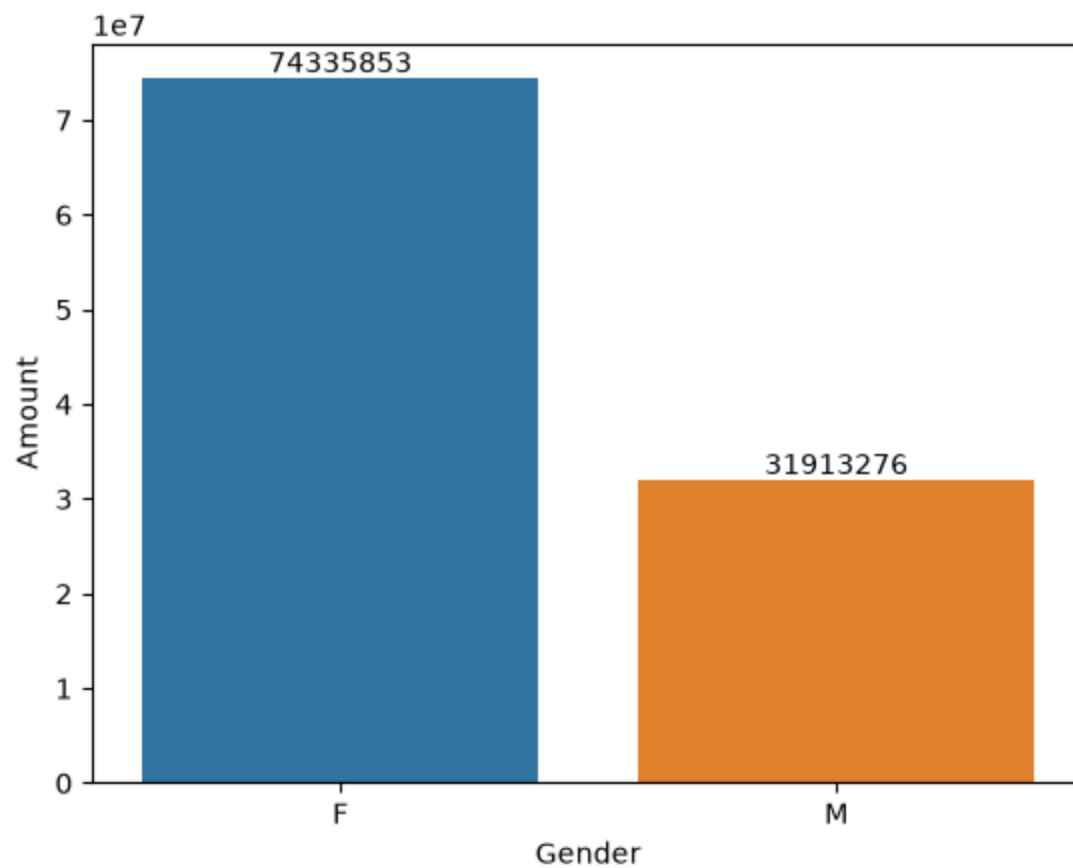
```
In [21]: sales_per_gender = df.groupby('Gender', as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)
```

```
In [22]: sales_per_gender
```

```
Out[22]:
```

	Gender	Amount
0	F	74335853
1	M	31913276

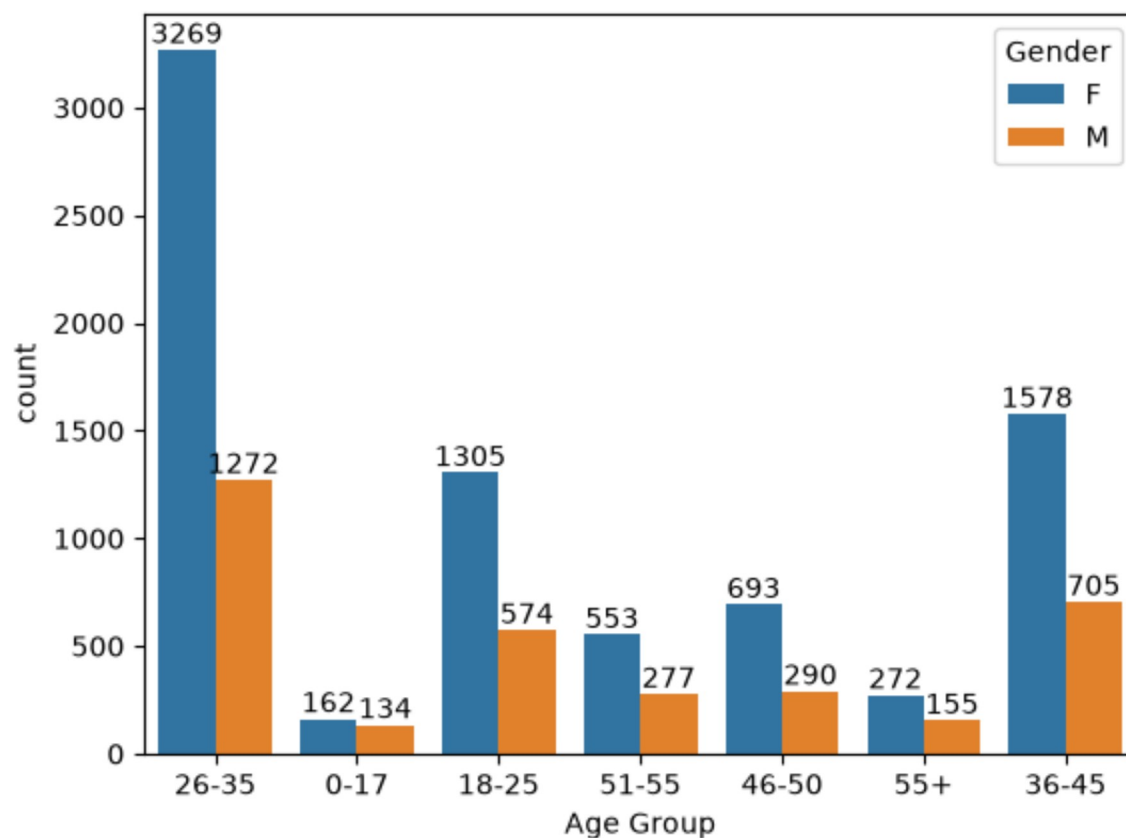
```
In [23]: ax = sns.barplot(data=sales_per_gender, x='Gender', y='Amount', hue='Gender')
for bars in ax.containers:
    ax.bar_label(bars, fmt='%.0f')
```



Age Wise Distribution

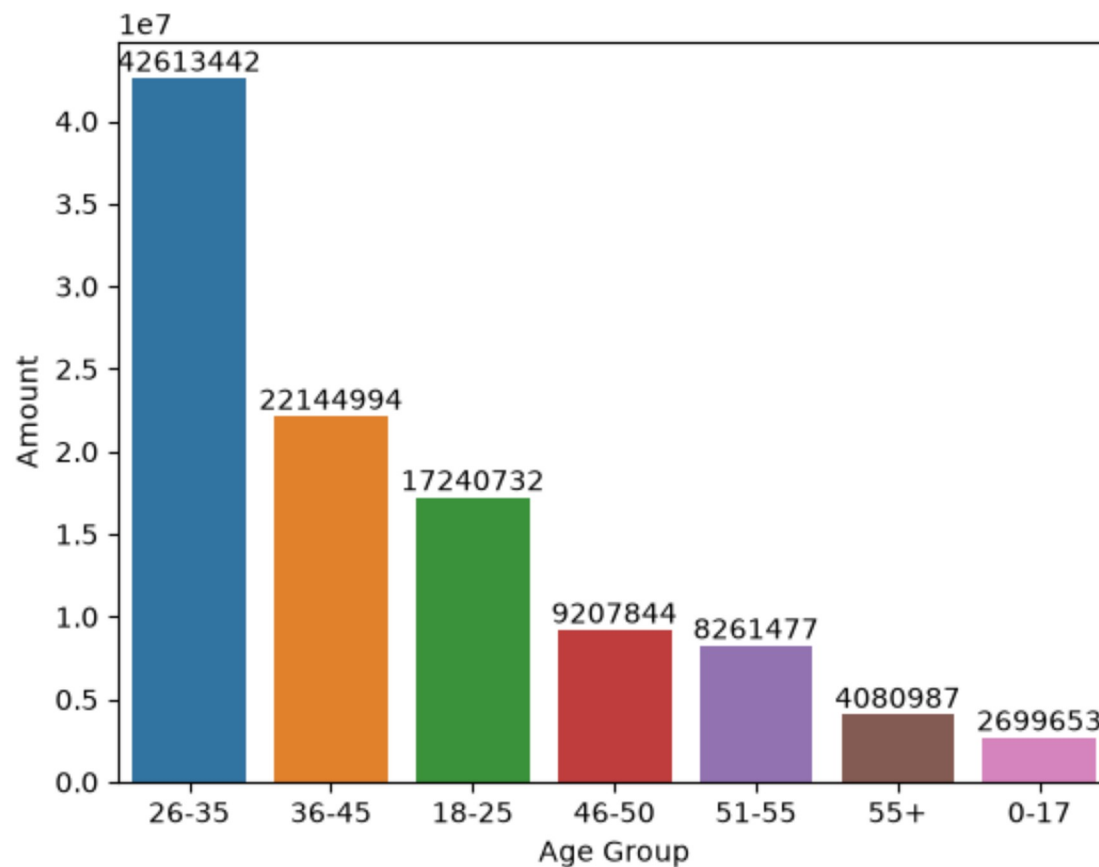
- Insights : Most of the buyers are from Age Group 26-35 and mostly are females.

```
In [24]: ax = sns.countplot(data=df, x='Age Group', hue='Gender')
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [25]: sales_per_age = df.groupby('Age Group', as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

ax = sns.barplot(data=sales_per_age, x='Age Group', y='Amount', hue='Age Group')
for bars in ax.containers:
    ax.bar_label(bars, fmt='%.0f')
```

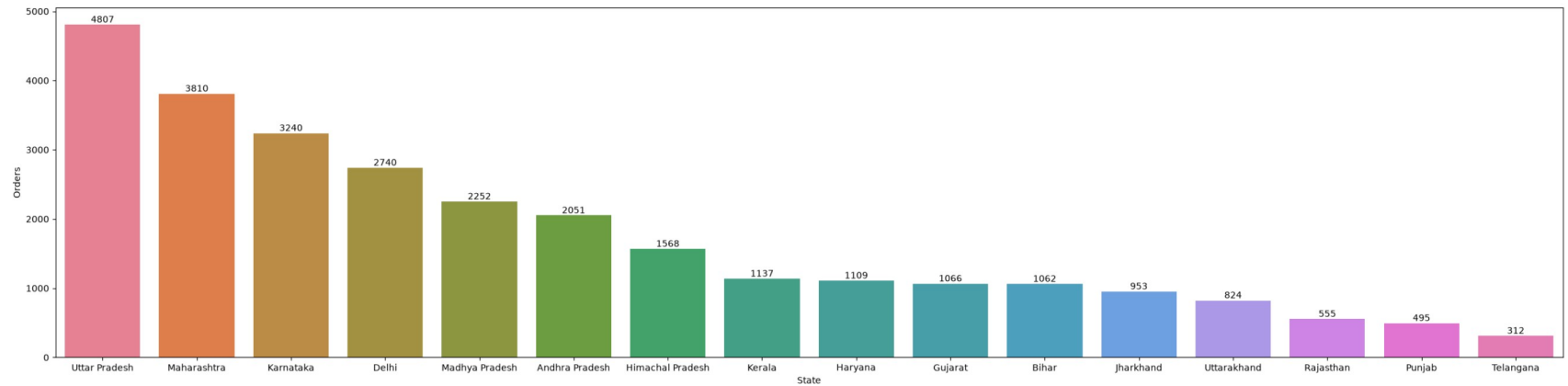


State Wise Total Orders

- Insights : Most of the orders are from Uttar Pradesh, Maharashtra and Karnataka.

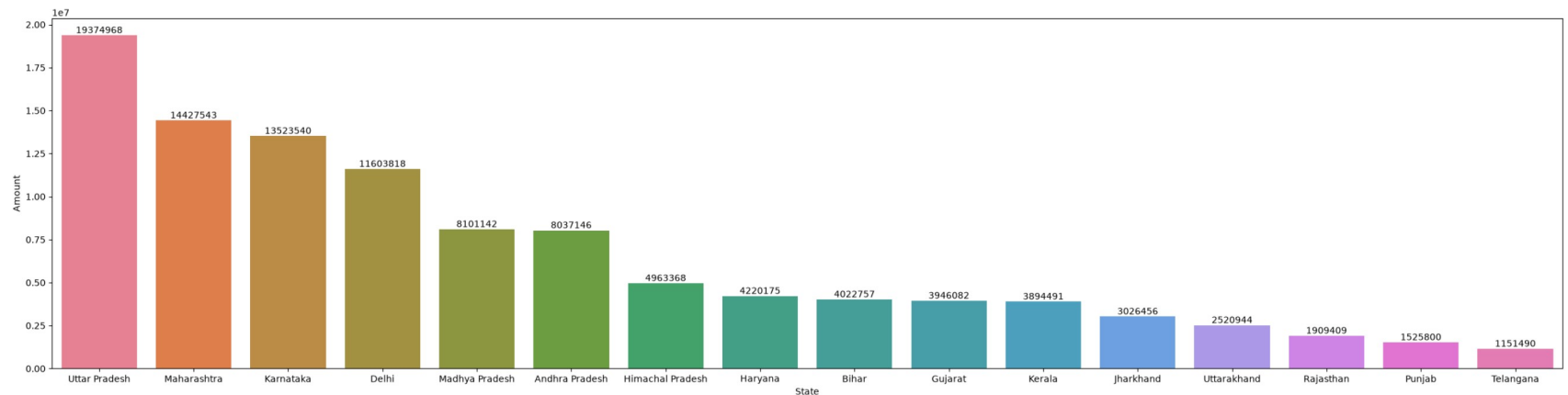
```
In [26]: state_wise_order = df.groupby('State', as_index=False)['Orders'].sum().sort_values(by='Orders', ascending=False)
```

```
In [27]: plt.figure(figsize=(30,7))
ax = sns.barplot(data=state_wise_order, x='State', y='Orders', hue='State')
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [28]: state_wise_amount = df.groupby('State', as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)
```

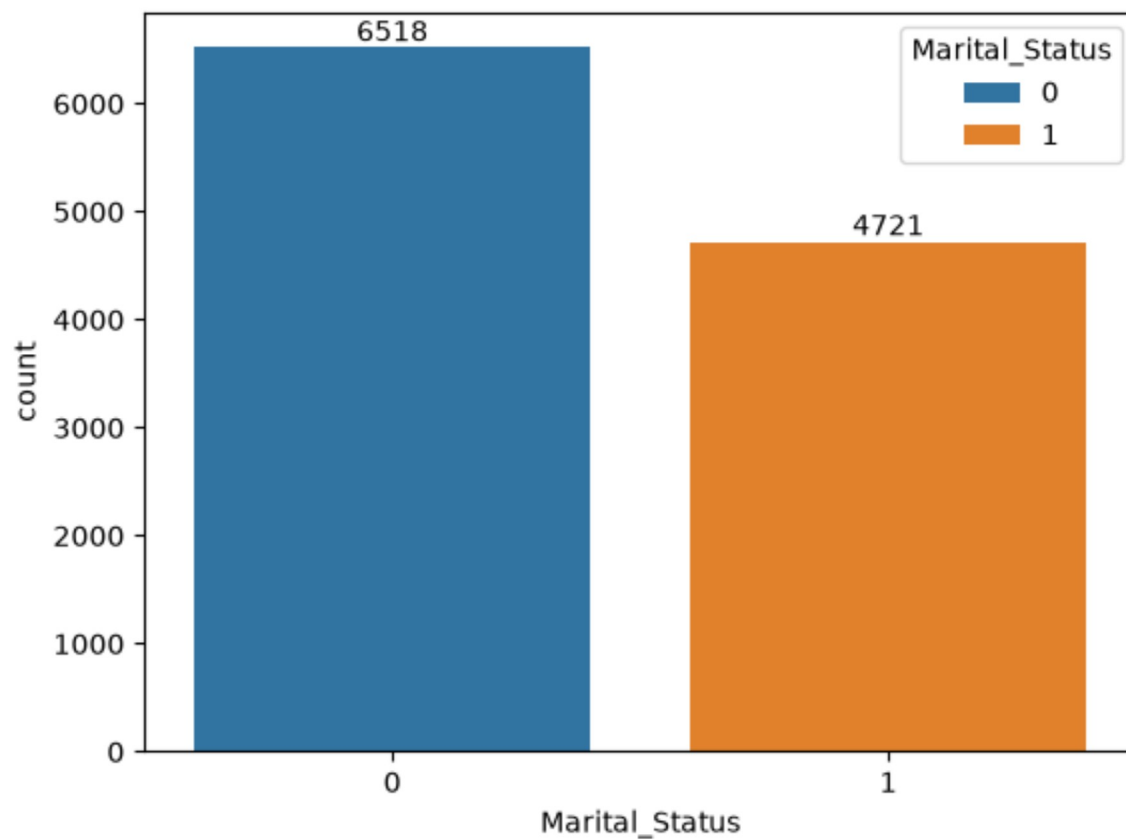
```
In [29]: plt.figure(figsize=(30,7))
ax = sns.barplot(data=state_wise_amount, x='State', y='Amount', hue='State')
for bars in ax.containers:
    ax.bar_label(bars, fmt='%.0f')
```



Marital Status

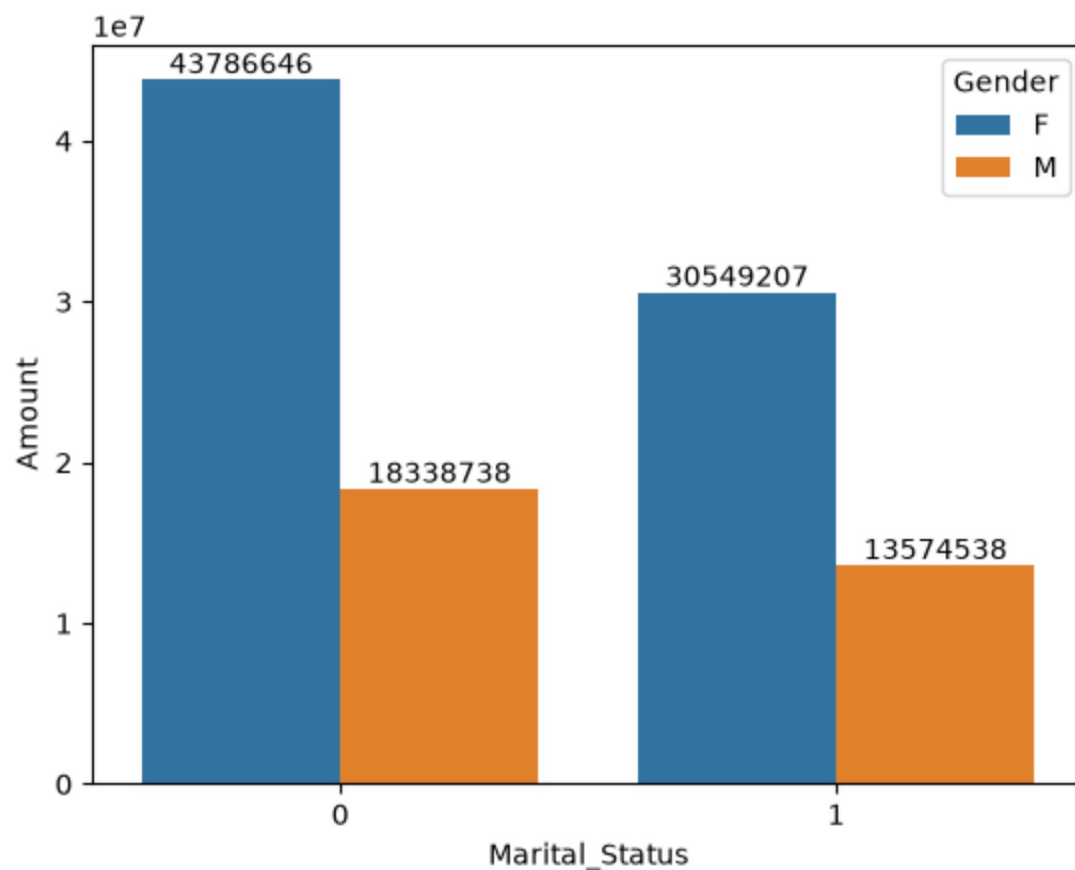
- Insights : Most of the buyers are married women and they have high purchasing power

```
In [30]: ax = sns.countplot(data=df, x='Marital_Status', hue='Marital_Status')
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [31]: marital_status_amount = df.groupby(['Marital_Status', 'Gender'], as_index=False)['Amount'].sum().sort_values(by='Amount')

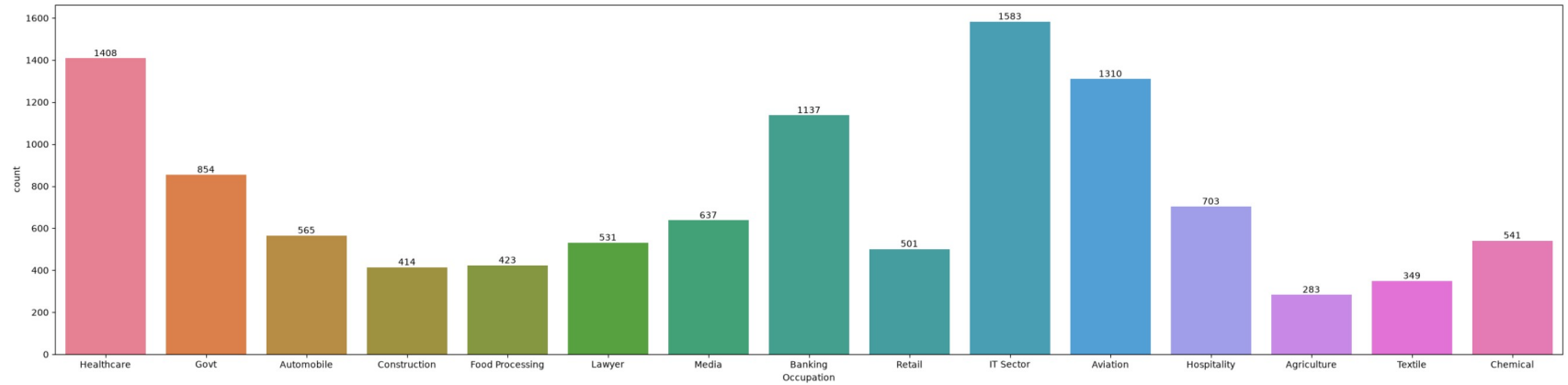
ax = sns.barplot(data=marital_status_amount, x='Marital_Status', y='Amount', hue='Gender')
for bars in ax.containers:
    ax.bar_label(bars, fmt='%.0f')
```



Occupation wise Distribution

- Insights : Most of the buyers are from IT Sector, Aviation and Healthcare sector.

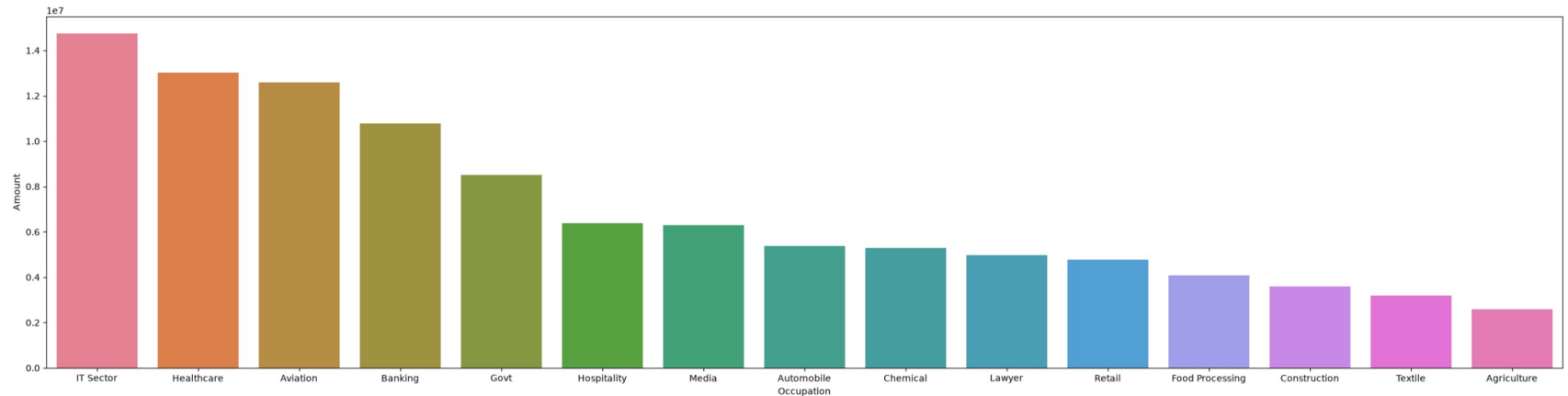
```
In [32]: plt.figure(figsize=(30,7))
ax = sns.countplot(data=df, x='Occupation', hue='Occupation')
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [33]: occupation_amount = df.groupby(['Occupation'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

plt.figure(figsize=(30,7))
sns.barplot(data=occupation_amount, x='Occupation', y='Amount', hue='Occupation')
```

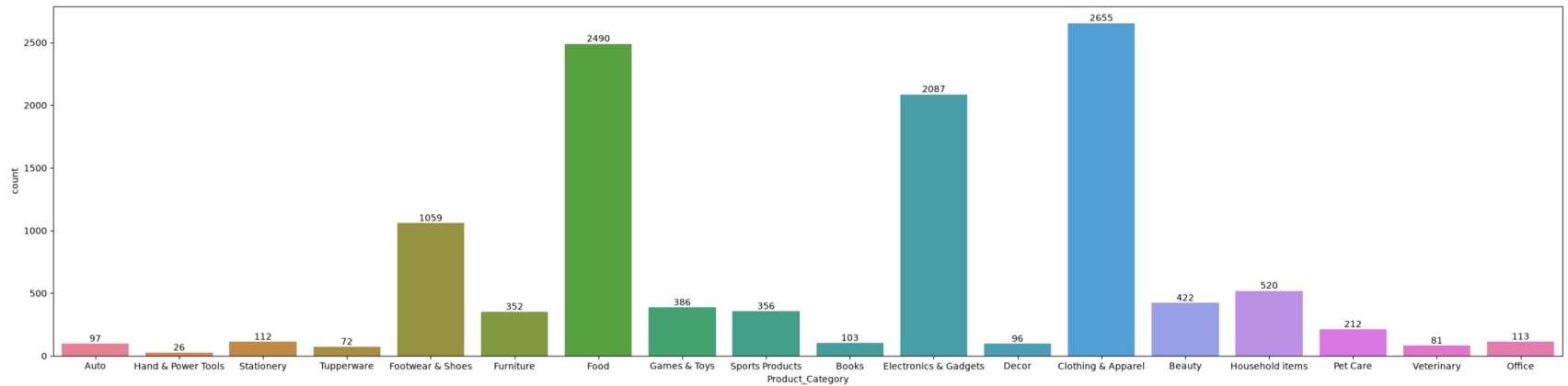
Out[33]: <Axes: xlabel='Occupation', ylabel='Amount'>



Product Category

- Insights - Most sold products are from food, clothing and Electronics

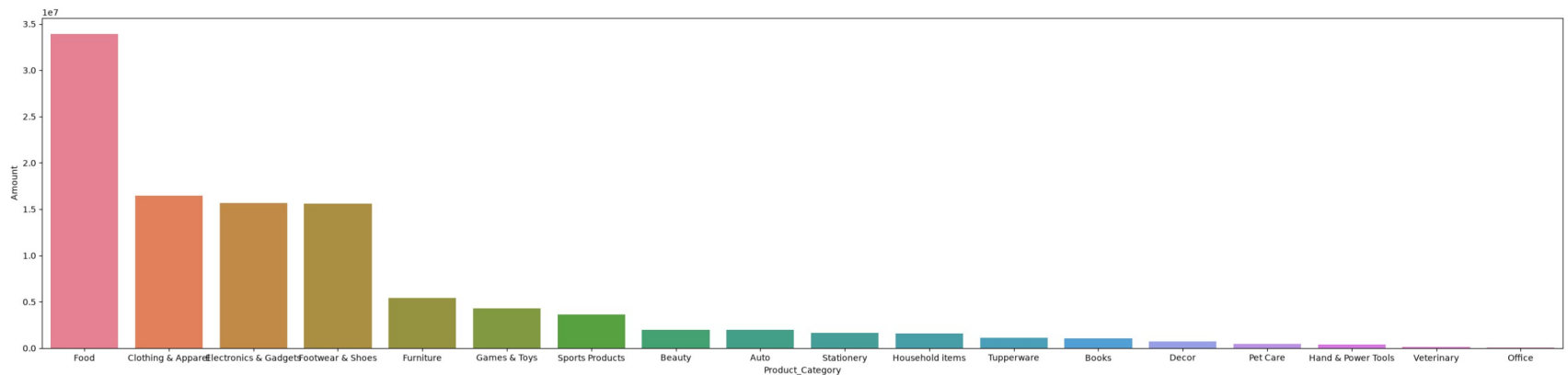
```
In [34]: plt.figure(figsize=(30,7))
ax = sns.countplot(data=df, x='Product_Category', hue='Product_Category')
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [35]: prd_cat_amount = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

plt.figure(figsize=(32,7))
sns.barplot(data=prd_cat_amount, x='Product_Category', y='Amount', hue='Product_Category')
```

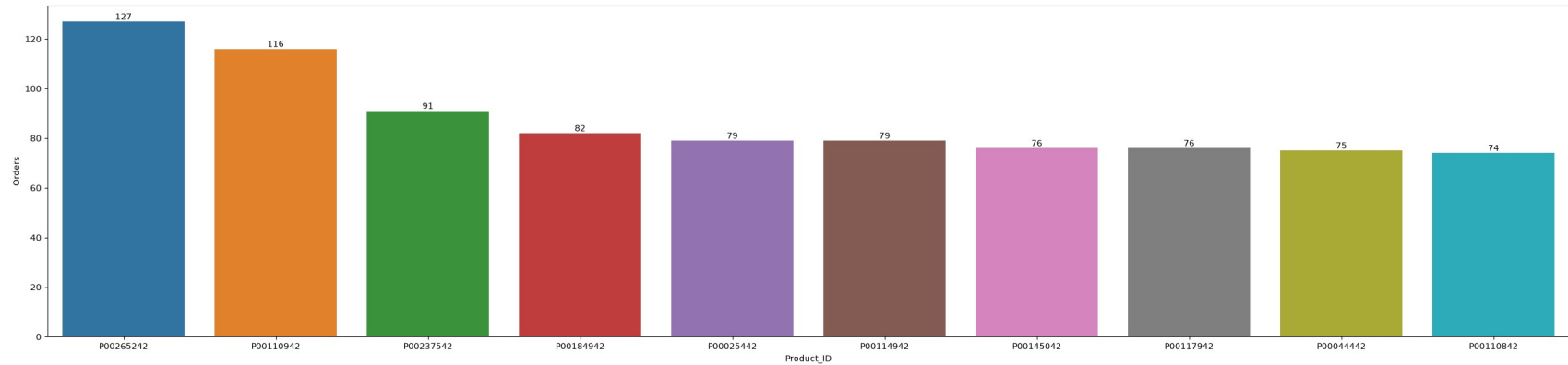
Out[35]: <Axes: xlabel='Product_Category', ylabel='Amount'>



```
In [37]: prd_ord = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_values(by='Orders', ascending=False).head(1)

plt.figure(figsize=(32,7))
ax = sns.barplot(data=prd_ord, x='Product_ID', y='Orders', hue='Product_ID')
```

```
for bars in ax.containers:  
    ax.bar_label(bars)
```



Conclusion

Married Women age group 26-35 yrs from UP, Maharashtra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from category Food, clothing and Electronics.